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Lap Timer Box

Version 2 - centred M18 sensor at 45 mm

1 Main shell: 69.1 X x 70.3 Y x 61.0 Z

2 Sensor axis: Z 45.0; M18 x 1; panel hole 19.0

3 Bottom tray seam: Z 11.0

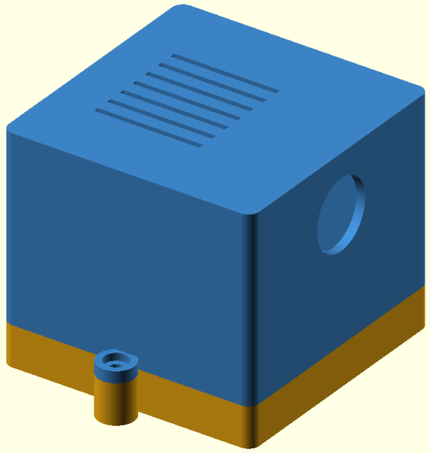
4 Main walls/floor: 2.4; roof: 2.0

5 PCB: 60.3 x 59.5 x 1.6; scan-derived mounting centres

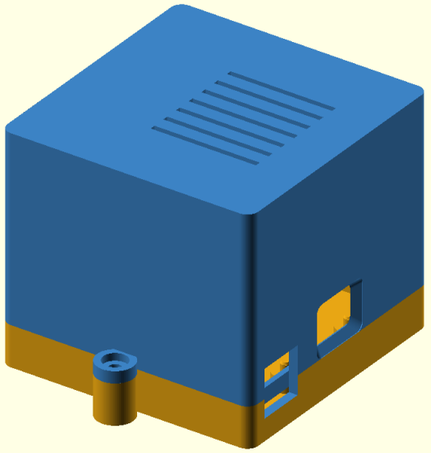
6 PCB clearance: rear 1.0; left/right/front 3.0

7 Tapered 2.8 locating pegs and rear passages are PROVISIONAL

8 Print base floor-down and supplied top roof-down



FRONT AND SENSOR SIDE



REAR CONNECTOR SIDE

VERSION 2 PURPOSE

Blue top and yellow bottom are separate parts.

The sensor is omitted so its hole is visible.

Print, fit and measure the real parts.

Record every mismatch for version 3.

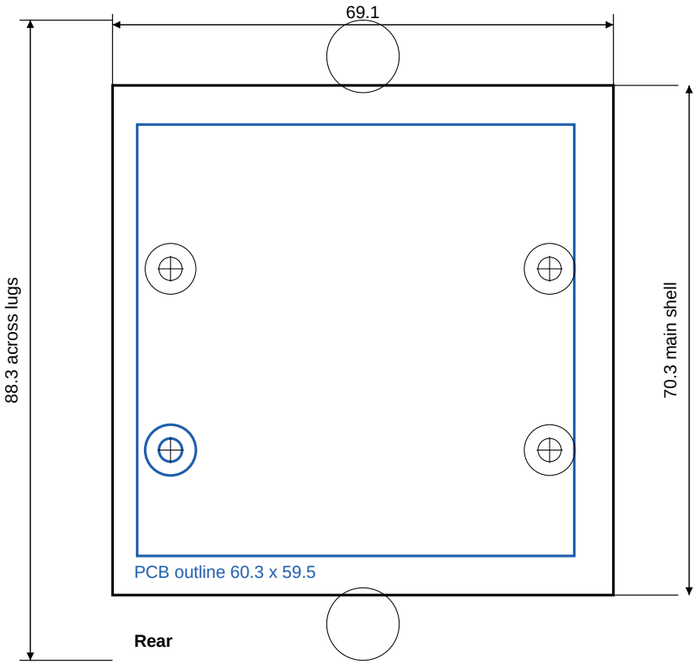
CAD nominal dimensions - unverified hardware interfaces are marked PROVISIONAL

ENCLOSURE AXONOMETRY

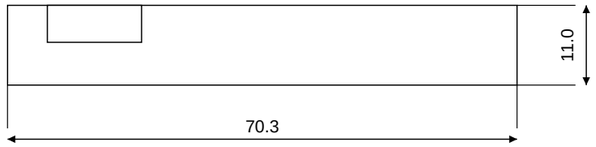
RoboSTEAM Lap Timer - version 2 prototype

Units: mm Scale: NTS Sheet 1/4

TOP VIEW



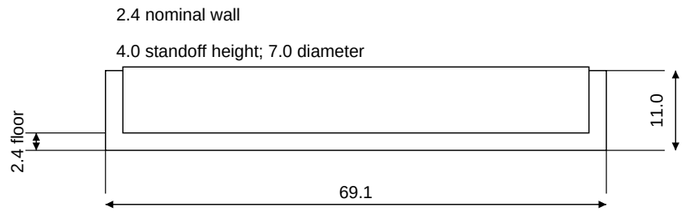
REAR VIEW



DC opening 13 x 13; centre 12.0 from rear-view left, Z 12.4 - PROVISIONAL

USB opening begins at/above seam; see top rear view

SECTION A-A THROUGH MAIN SHELL



PCB mounting centres from PCB rear-left origin

(4.6, 14.6) (4.6, 39.6) (56.9, 14.6) (56.9, 39.6)

Global PCB origin in tray: X 3.4, Y 5.4. Tapered pin diameter 2.8 is PROVISIONAL.

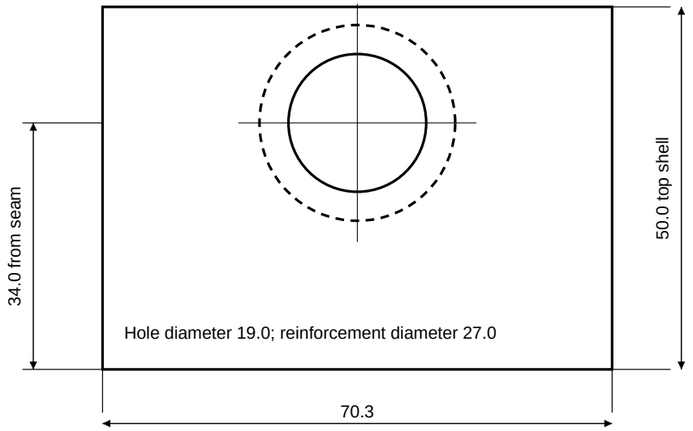
CAD nominal dimensions - unverified hardware interfaces are marked PROVISIONAL

BOTTOM TRAY - ORTHOGRAPHIC VIEWS AND SECTION

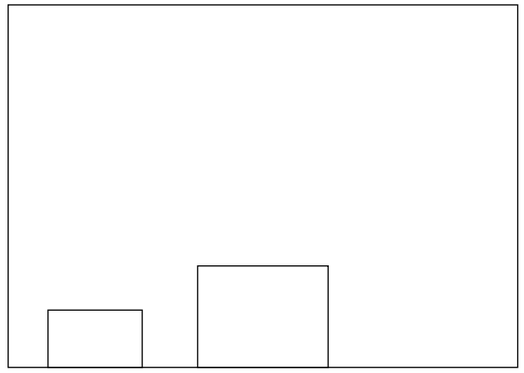
RoboSTEAM Lap Timer - version 2 prototype

Units: mm Scale: 1:1.25 Sheet 2/4

FRONT VIEW



REAR VIEW - OUTSIDE FACE



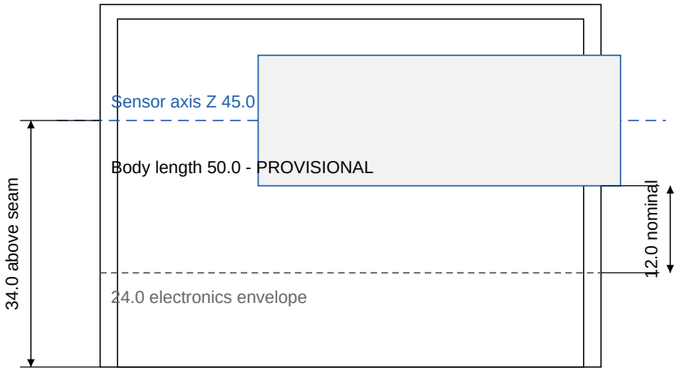
USB outer flare 18 x 14

USB inner 14 x 10

From rear-view left: USB 35.15; DC 12.0. Heights Z 18.0 and 12.4.

All rear opening dimensions are PROVISIONAL

SECTION B-B THROUGH SENSOR AXIS



TOP PRINT ORIENTATION

Supplied STL places the 2.0 roof on the build plate and the locating lip upward.

Locating lip: height 4.0; thickness 1.4; mating clearance 0.30 per side.

Closure: 3.4 clearance, 6.5 head recess, 2.6 base pilot - PROVISIONAL.

CAD nominal dimensions - unverified hardware interfaces are marked PROVISIONAL

REMOVABLE TOP - VIEWS, OPENINGS AND SENSOR SECTION

RoboSTEAM Lap Timer - version 2 prototype

Units: mm Scale: 1:1.25 Sheet 3/4

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STEAM
ACADEMY

ABOUT THIS DOCUMENT

RoboSTEAM Lap Timer - a result of the Erasmus+ project ROBO STEAM ACADEMY (KA220-VET - Cooperation partnerships in vocational education and training, project number KA220-VET-7CF4F308, 2024-2027), used for the project's robot competitions (Work Package 3). Project website: https://robosteam.eu/

PARTNERS

National STEM Center (coordinator, Bulgaria); Institute of Information and Communication Technologies, Bulgarian Academy of Sciences (Bulgaria); Academy Nikola Tesla (Bulgaria); Technical University of Kosice (Slovakia); SPSE Presov (Slovakia); National Association of Public Librarians and Libraries in Romania - ANBPR (Romania); Liceul Tehnologic Cezar Nicolau (Romania).

LICENCE

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RoboSTEAM Lap Timer - version 2 prototype

Units: mm Scale: NTS Sheet 4/4

CAD nominal dimensions - unverified hardware interfaces are marked PROVISIONAL